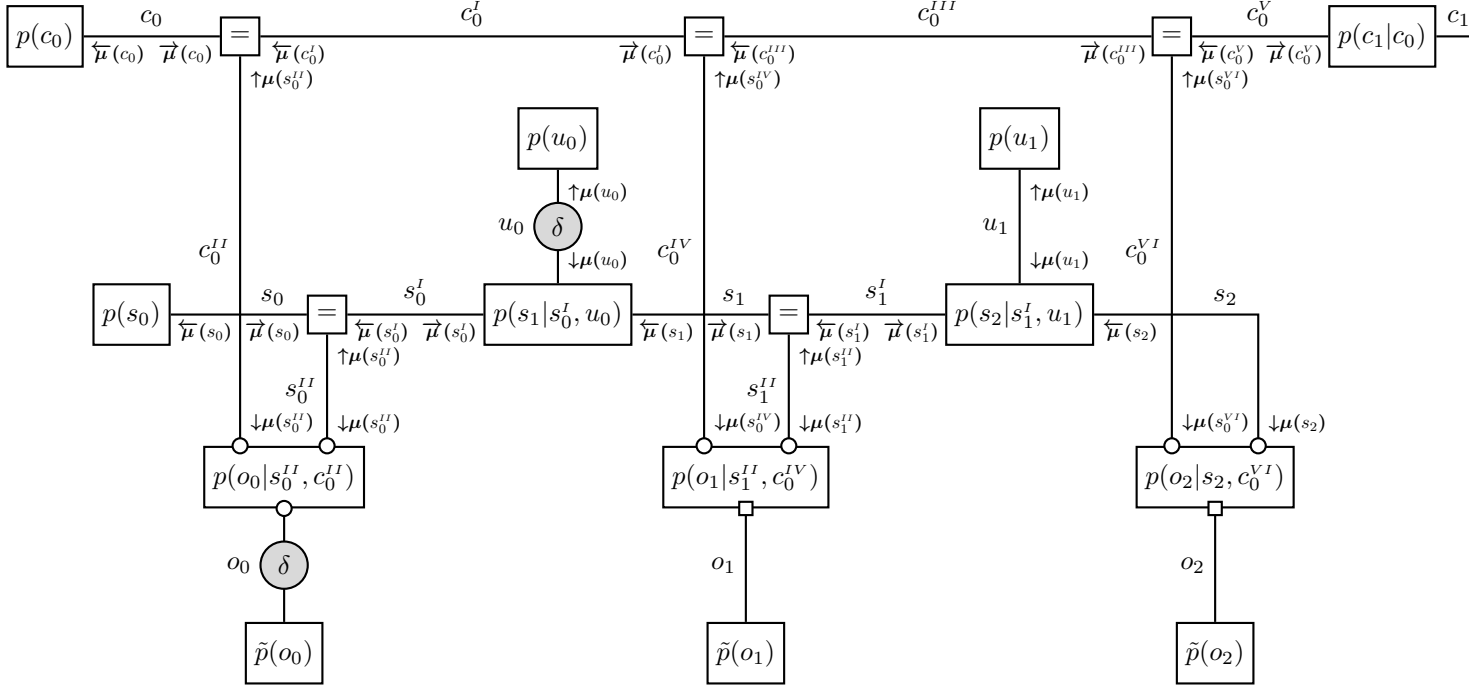


Setup Context GFE (current timestep $t = 1$)



Bethe Forward Sweep (s and u):

$$\vec{\mu}(s_0) = p(s_0) \quad (1)$$

$$\uparrow\mu(s_0^{II}) = \exp\left(\sum_{o_0, c_0^{II}} q(o_0)q(c_0^{II}) \log p(o_0|s_0^{II}, c_0^{II})\right) \quad (2)$$

$$\vec{\mu}(s_0^I) = \vec{\mu}(s_0) \uparrow\mu(s_0^{II}) \quad (3)$$

$$\downarrow\mu(u_0) = \delta_{u_0, \hat{u}_o} \quad (4)$$

$$\vec{\mu}(s_1) = \sum_{s_0^I, u_0} \vec{\mu}(s_0^I) \downarrow\mu(u_0) p(s_1|s_0^I, u_0) \quad (5)$$

$$q^{(k)}(o_1; \nu) = \sum_{s_1^{II}, c_0^{IV}} q^{(k-1)}(s_1^{II}; \nu) q(c_0^{IV}) p(o_1|s_1^{II}, c_0^{IV}) \quad (6)$$

$$\tilde{f}^{(k)}(s_1^{II}; \nu) = \exp\left(\sum_{o_1, c_0^{IV}} p(o_1|s_1^{II}, c_0^{IV}) q(c_0^{IV}) \log \frac{p(o_1|s_1^{II}, c_0^{IV}) \tilde{p}(o_1)}{q^{(k)}(o_1; \nu)}\right)$$

$$q^{(k)}(s_1^{II}; \nu) = \frac{\tilde{f}^{(k)}(s_1^{II}; \nu) \downarrow\mu(s_1^{II})}{\sum_{s_1^{II}} \tilde{f}^{(k)}(s_1^{II}; \nu) \downarrow\mu(s_1^{II})}$$

$$\uparrow\mu(s_1^{II}) = q(s_1^{II}; \nu^*) / \downarrow\mu(s_1^{II}) \quad (7)$$

$$\vec{\mu}(s_1^I) = \vec{\mu}(s_1) \uparrow\mu(s_1^{II}) \quad (8)$$

$$\downarrow\mu(u_1) = p(u_1) \quad (9)$$

$$\downarrow\mu(s_2) = \sum_{s_1^I, u_1} \vec{\mu}(s_1^I) \downarrow\mu(u_1) p(s_2|s_1^I, u_1) \quad (10)$$

Bethe Backward Sweep (s and u):

$$q^{(k)}(o_2; \nu) = \sum_{s_2, c_0^{VI}} q^{(k-1)}(s_2; \nu) q(c_0^{VI}) p(o_2|s_2, c_0^{VI}) \quad (11)$$

$$\tilde{f}^{(k)}(s_2; \nu) = \exp\left(\sum_{o_2, c_0^{VI}} p(o_2|s_2, c_0^{VI}) q(c_0^{VI}) \log \frac{p(o_2|s_2, c_0^{VI}) \tilde{p}(o_2)}{q^{(k)}(o_2; \nu)}\right)$$

$$q^{(k)}(s_2; \nu) = \frac{\tilde{f}^{(k)}(s_2; \nu) \downarrow\mu(s_2)}{\sum_{s_2} \tilde{f}^{(k)}(s_2; \nu) \downarrow\mu(s_2)}$$

$$\vec{\mu}(s_2) = q(s_2; \nu^*) / \downarrow\mu(s_2) \quad (12)$$

$$\uparrow\mu(u_1) = \sum_{s_1^I, s_2} \vec{\mu}(s_1^I) \check{\mu}(s_2) p(s_2 | s_1^I, u_1) \quad (13)$$

$$\check{\mu}(s_1^I) = \sum_{u_1, s_2} \downarrow\mu(u_1) \check{\mu}(s_2) p(s_2 | s_1^I, u_1) \quad (14)$$

$$\downarrow\mu(s_1^{II}) = \vec{\mu}(s_1) \check{\mu}(s_1^I) \quad (15)$$

$$\downarrow\mu(o_1) = \sum_{s_1^{II}} \tilde{f}(o_1, s_1^{II}) \downarrow\mu(s_1^{II}) \quad (16)$$

$$\check{\mu}(s_1) = \uparrow\mu(s_1^{II}) \check{\mu}(s_1^I) \quad (17)$$

$$\uparrow\mu(u_0) = \delta_{u_0, \hat{u}_0} \quad (18)$$

$$\check{\mu}(s_0^I) = \sum_{u_0, s_1} \downarrow\mu(u_0) \check{\mu}(s_1) p(s_1 | s_0^I, u_0) \quad (19)$$

$$\downarrow\mu(s_0^{II}) = \vec{\mu}(s_0) \check{\mu}(s_0^I) \quad (20)$$

$$\check{\mu}(s_0) = \uparrow\mu(s_0^{II}) \check{\mu}(s_0^I) \quad (21)$$

Updating Beliefs (s, u and o):

$$q(s_0) \propto \check{\mu}(s_0) \vec{\mu}(s_0) \quad (22)$$

$$q(s_0^I) \propto \check{\mu}(s_0^I) \vec{\mu}(s_0^I) \quad (23)$$

$$q(s_0^{II}) \propto \uparrow\mu(s_0^{II}) \downarrow\mu(s_0^{II}) \quad (24)$$

$$q(s_1) \propto \check{\mu}(s_1) \vec{\mu}(s_1) \quad (25)$$

$$q(s_1^I) \propto \check{\mu}(s_1^I) \vec{\mu}(s_1^I) \quad (26)$$

$$q(s_1^{II}) \propto \uparrow\mu(s_1^{II}) \downarrow\mu(s_1^{II}) \quad (27)$$

$$q(s_2) \propto \check{\mu}(s_2) \downarrow\mu(s_2) \quad (28)$$

$$q(u_0) \propto \delta_{u_0, \hat{u}_0} \quad (29)$$

$$q(u_1) \propto \uparrow\mu(u_1) \downarrow\mu(u_1) \quad (30)$$

$$q(o_0) \propto \delta_{o_0, \hat{o}_0} \quad (31)$$

$$q(o_1) \propto \sum_{c_0^{IV}, s_1^{II}} q(c_0^{IV}) q(s_1^{II}) p(o_1 | s_1^{II}, c_0^{IV}) \quad (32)$$

$$q(o_2) \propto \sum_{c_0^{VI}, s_2} q(c_0^{VI}) q(s_2) p(o_2 | s_2, c_0^{VI}) \quad (33)$$

Bethe Forward Sweep (c):

$$\vec{\mu}(c_0) = p(c_0) \quad (34)$$

$$\uparrow\mu(c_0^{II}) = \exp\left(\sum_{o_0, s_0^{II}} q(o_0, s_0^{II}) \log p(o_0 | s_0^{II}, c_0^{II})\right) \quad (35)$$

$$\vec{\mu}(c_0^I) = \vec{\mu}(c_0) \uparrow\mu(c_0^{II}) \quad (36)$$

$$q^{(k)}(o_1; \nu) = \sum_{s_1^{II}, c_0^{IV}} q^{(k-1)}(s_1^{II}) q(c_0^{IV}; \nu) p(o_1 | s_1^{II}, c_0^{IV}) \quad (37)$$

$$\tilde{f}^{(k)}(c_0^{IV}; \nu) = \exp\left(\sum_{o_1, s_1^{II}} p(o_1 | s_1^{II}, c_0^{IV}) q(s_1^{II}) \log \frac{p(o_1 | s_1^{II}, c_0^{IV}) \tilde{p}(o_1)}{q^{(k)}(o_1; \nu)}\right)$$

$$q^{(k)}(c_0^{IV}; \nu) = \frac{\tilde{f}^{(k)}(c_0^{IV}; \nu) \downarrow\mu(c_0^{IV})}{\sum_{c_0^{IV}} \tilde{f}^{(k)}(c_0^{IV}; \nu) \downarrow\mu(c_0^{IV})}$$

$$\uparrow\mu(c_0^{IV}) = q(c_0^{IV}; \nu^*) / \downarrow\mu(c_0^{IV}) \quad (38)$$

$$\vec{\mu}(c_0^{III}) = \vec{\mu}(c_0^I) \uparrow\mu(c_0^{IV}) \quad (39)$$

$$q^{(k)}(o_2; \nu) = \sum_{s_2^{II}, c_0^{VI}} q^{(k-1)}(s_2^{II}) q(c_0^{VI}; \nu) p(o_2 | s_2^{II}, c_0^{VI}) \quad (40)$$

$$\tilde{f}^{(k)}(c_0^{VI}; \nu) = \exp\left(\sum_{o_2, s_2^{II}} p(o_2 | s_2^{II}, c_0^{VI}) q(s_2^{II}) \log \frac{p(o_2 | s_2^{II}, c_0^{VI}) \tilde{p}(o_2)}{q^{(k)}(o_2; \nu)}\right)$$

$$q^{(k)}(c_0^{VI}; \nu) = \frac{\tilde{f}^{(k)}(c_0^{VI}; \nu) \downarrow\mu(c_0^{VI})}{\sum_{c_0^{VI}} \tilde{f}^{(k)}(c_0^{VI}; \nu) \downarrow\mu(c_0^{VI})}$$

$$\uparrow\mu(c_0^{VI}) = q(c_0^{VI}; \nu^*) / \downarrow\mu(c_0^{VI}) \quad (41)$$

$$\vec{\mu}(c_0^V) = \vec{\mu}(c_0^{III}) \uparrow\mu(c_0^{VI}) \quad (42)$$

Bethe Backward Sweep (c):

$$\check{\mu}(c_0^V) = [0.5, 0.5]^T \quad (43)$$

$$\downarrow\mu(c_0^{VI}) = \vec{\mu}(c_0^{III}) \check{\mu}(c_0^V) \quad (44)$$

$$\check{\mu}(c_0^{III}) = \uparrow\mu(c_0^{VI}) \check{\mu}(c_0^V) \quad (45)$$

$$\downarrow\mu(c_0^{IV}) = \vec{\mu}(c_0^I) \check{\mu}(c_0^{III}) \quad (46)$$

$$\check{\mu}(c_0^I) = \uparrow\mu(c_0^{IV}) \check{\mu}(c_0^{III}) \quad (47)$$

$$\downarrow\mu(c_0^{II}) = \vec{\mu}(c_0) \check{\mu}(c_0^I) \quad (48)$$

$$\check{\mu}(c_0) = \uparrow\mu(c_0^{II}) \check{\mu}(c_0^I) \quad (49)$$

$$(50)$$

Updating Beliefs (c):

$$q(c_0) \propto \check{\mu}(c_0) \vec{\mu}(c_0) \quad (51)$$

$$q(c_0^I) \propto \check{\mu}(c_0^I) \vec{\mu}(c_0^I) \quad (52)$$

$$q(c_0^{II}) \propto \uparrow\mu(c_0^{II}) \downarrow\mu(c_0^{II}) \quad (53)$$

$$q(c_0^{III}) \propto \check{\mu}(c_0^{III}) \vec{\mu}(c_0^{III}) \quad (54)$$

$$q(c_0^{IV}) \propto \uparrow\mu(c_0^{IV}) \downarrow\mu(c_0^{IV}) \quad (55)$$

$$q(c_0^V) \propto \check{\mu}(c_0^V) \vec{\mu}(c_0^V) \quad (56)$$

$$q(c_0^{VI}) \propto \uparrow\mu(c_0^{VI}) \downarrow\mu(c_0^{VI}) \quad (57)$$

Repeat Eq.1-57 until convergence!

$$\begin{aligned}
F = & \sum_{s_0} q(s_0) \log \frac{q(s_0)}{p(s_0)} + \sum_{s_1, s_0^I, u_0} q(s_1, s_0^I, u_0) \log \frac{q(s_1, s_0^I, u_0)}{p(s_1|s_0^I, u_0)} + \sum_{s_2, s_1^I, u_1} q(s_2, s_1^I, u_1) \log \frac{q(s_2, s_1^I, u_1)}{p(s_2|s_1^I, u_1)} \\
& + \sum_{u_0} q(u_0) \log \frac{q(u_0)}{p(u_0)} + \sum_{u_1} q(u_1) \log \frac{q(u_1)}{p(u_1)} \\
& + \sum_{o_0, s_0^{II}, c_0^{II}} q(o_0) q(s_0^{II}) q(c_0^{II}) \log \frac{q(o_0) q(s_0^{II}) q(c_0^{II})}{p(o_0|s_0^{II}, c_0^{II})} + \sum_{o_1, s_1^{II}, c_0^{IV}} p(o_1|s_1^{II}, c_0^{IV}) q(s_1^{II}) q(c_0^{IV}) \log \frac{q(o_1) q(s_1^{II}) q(c_0^{IV})}{p(o_1|s_1^{II}, c_0^{IV})} \\
& + \sum_{o_2, s_2, c_0^{VI}} p(o_2|s_2, c_0^{VI}) q(s_2) q(c_0^{VI}) \log \frac{q(o_2) q(s_2) q(c_0^{VI})}{p(o_2|s_2, c_0^{VI})} \\
& + \sum_{o_0} q(o_0) \log \frac{q(o_0)}{\tilde{p}(o_0)} + \sum_{o_1} q(o_1) \log \frac{q(o_1)}{\tilde{p}(o_1)} + \sum_{o_2} q(o_2) \log \frac{q(o_2)}{\tilde{p}(o_2)} \\
& + \sum_{s_0, s_0^I, s_0^{II}} q(s_0, s_0^I, s_0^{II}) \log \frac{q(s_0, s_0^I, s_0^{II})}{\delta_{s_0, s_0^I} \delta_{s_0, s_0^{II}}} + \sum_{s_1, s_1^I, s_1^{II}} q(s_1, s_1^I, s_1^{II}) \log \frac{q(s_1, s_1^I, s_1^{II})}{\delta_{s_1, s_1^I} \delta_{s_1, s_1^{II}}} \\
& + \sum_{c_0} q(c_0) \log \frac{q(c_0)}{p(c_0)} \\
& + \sum_{c_0, c_0^I, c_0^{II}} q(c_0, c_0^I, c_0^{II}) \log \frac{q(c_0, c_0^I, c_0^{II})}{\delta_{c_0, c_0^I} \delta_{c_0, c_0^{II}}} + \sum_{c_0^I, c_0^{III}, c_0^{IV}} q(c_0^I, c_0^{III}, c_0^{IV}) \log \frac{q(c_0^I, c_0^{III}, c_0^{IV})}{\delta_{c_0^I, c_0^{III}} \delta_{c_0^I, c_0^{IV}}} \\
& + \sum_{c_0^{III}, c_0^V, c_0^{VI}} q(c_0^{III}, c_0^V, c_0^{VI}) \log \frac{q(c_0^{III}, c_0^V, c_0^{VI})}{\delta_{c_0^{III}, c_0^V} \delta_{c_0^{III}, c_0^{VI}}} \\
& - \sum_{s_0} q(s_0) \log q(s_0) - \sum_{s_0^I} q(s_0^I) \log q(s_0^I) - \sum_{s_0^{II}} q(s_0^{II}) \log q(s_0^{II}) \\
& - \sum_{s_1} q(s_1) \log q(s_1) - \sum_{s_1^I} q(s_1^I) \log q(s_1^I) - \sum_{s_1^{II}} q(s_1^{II}) \log q(s_1^{II}) - \sum_{s_2} q(s_2) \log q(s_2) \\
& - \sum_{u_0} q(u_0) \log q(u_0) - \sum_{u_1} q(u_1) \log q(u_1) \\
& - \sum_{o_0} q(o_0) \log q(o_0) - \sum_{o_1} q(o_1) \log q(o_1) - \sum_{o_2} q(o_2) \log q(o_2) \\
& - \sum_{c_0} q(c_0) \log q(c_0) - \sum_{c_0^I} q(c_0^I) \log q(c_0^I) - \sum_{c_0^{II}} q(c_0^{II}) \log q(c_0^{II}) - \sum_{c_0^{III}} q(c_0^{III}) \log q(c_0^{III}) \\
& - \sum_{c_0^{IV}} q(c_0^{IV}) \log q(c_0^{IV}) - \sum_{c_0^{VI}} q(c_0^{VI}) \log q(c_0^{VI})
\end{aligned}$$